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Research Keywords

Metal-Organic Frameworks (MOFs), Organic Synthesis, Reticular Chemistry, Functional Porous Materials, Heterogeneous Catalyst, Gas Sorption & Separation, Water sorption, Heterogeneous Catalyst, Photophysical Property, Electrocatalysis, Bioinspired Materials, Data-Driven Materials Discovery, Water electrolysis

Professional Career

Postdoctoral Associate, Nov. 2025 – Present / Advisor: Prof. Jungki Ryu
InnoCORE fellow
UNIST, Ulsan, Republic of Korea

Postdoctoral Associate, Nov. 2024 – Oct. 2025 / Advisor: Prof. Liang Feng
Thomas Lord Department of Mechanical Engineering & Materials Science
Duke University, Durham, NC, USA

Postdoctoral Associate, Mar. 2023 – Oct. 2024 / Advisor: Prof. Wonyoung Choe
Department of Chemistry
UNIST, Ulsan, Republic of Korea

Education

02/2023 **Ph.D. (Chemistry)**, Ulsan National Institute of Science and Technology (UNIST), Ulsan, 44919, Republic of Korea (Advisor: Prof. Wonyoung Choe)

Dissertation title: Rational Synthetic Strategy for Porphyrinic and Related Metal-Organic Frameworks Targeting Advanced Functionality

02/2015 **B.S. (Applied Chemistry)**, Kumoh National Institute of Technology, Gumi, Republic of Korea

Publications (†Co-first author, *Corresponding author)

1. Huang, L.;† **Lee, J.**;† Xiang, L.; Feng, L.*
Creating Compartmentalized Pockets as Biomimetic Synthesizer for Length-Tunable Peptides
***J. Am. Chem. Soc.* 2026**, *148*, 18531-18539.

2. **Lee, J.**; Xiang, L.; Huang, L.; Feng, L.*
Constructing Polyamine-Rich Zr-Based Metal-Organic Frameworks via Linker Vacancy Engineering for Carbon Capture
***ACS Materials Lett.* 2026**, *8*, 1872-1881.

3. **Lee, J.**;† Kim, K. M.;† Ga, S.; Oh, H.*; Ryu, J.*
Decoupled water electrolysis as an Integrated Hydrogen Carrier Platform
***Adv. Energy Mater.* 2026**, e71763.

4. **Lee, J.**; Park, D.; Jin, E.; Lee, S.; Lee, J.; Oh, H.;* Choe, W.*
Programmable Merged-Net Porphyrinic Metal-Organic Frameworks for Water Sorption
***Adv. Funct. Mater.* 2025**, 35, 2413200.

5. **Lee, J.**;† Hwang, E.;† Kim, T. I.; Jin, E.; Cho, E.; Park, Y. S.;* Min, S. K.;* Kwon, T.-H.;* and Choe, W.*
Unveiling Energy Transfer Mechanisms in Nanographene-incorporated Metal-Organic Frameworks
***ACS Materials Lett.* 2024**, *6*, 1151-1159.

6. **Lee, J.**;† Choi, H.;† Mun, J.; Jin, E.; Lee S.; Nam J.; Umer M.; Cho J.; Lee, G.;* Kwon, Y.;* Choe. W.*
Nanozyme Based on Porphyrinic Metal-Organic Framework for Electrocatalytic CO_2 Reduction
Small Struct.* 2023**, *4*, 2200087. ***(Selected as Inside Front Cover)

7. Kong, E.; Lee, Y. C.; Kim, H.; **Lee, J.**; Oh, T.;* Ryu, J.*
Carbon Monoxide Oxidation Enables Energy-Efficient Hydrogen Peroxide and Carbonate Co-Production from CO-Rich Flue Gas and Air
***ACS Energy Lett.* 2026**, *In press*

8. Zhang, P.; Xiang, L.; **Lee, J.**; Chen, Z.; Huang, L.; Feng, L.*
Temperature-Sensitive Carbon Capture in MOFs for Energy-Efficient Flue Gas and Biogas Purification
***ACS Appl. Mater. Interfaces* 2026**, *18*, 39259-39266.

9. Chen, Z.; Xiang, L.; Zhang, P.; **Lee, J.**; Huang, L.; Feng, L.*
Upcycling Plastic Waste into Hydrophobic-Shielded Adsorbents for Efficient Direct Air Capture
***ACS Appl. Mater. Interfaces* 2026**, *18*, 31292-31300.

10. Jin, E.; Lee, I. S.; Yang, D. C.; Moon, D.; Nam, J.; Cho, H.; Kang, E.; **Lee, J.**; Noh, H.-J.; Min, S. K.;* Choe, W.*
Origamic Metal-Organic Framework toward mechanical metamaterial
***Nat. Commun.* 2023**, *14*, 7938.

11. Chen, Z.; Xiang, L.;* Zhang, P.; Huang, L.; **Lee, J.**; Feng, L.*
Engineering Porous Materials for Real-World Carbon Capture: Mechanisms, Design, and Environmental Impact
Submitted, **2026**

12. Oh, H.;† Jeon, S.;† Kim, K. M.; **Lee, J.**; Lee, I.; Ryu, J.*
Ambient-air-driven electrochemical dechlorination of PVC for waste valorization
Under revision, **2026**

13. Xiang, L.;† **Lee, J.**;† Zhang, Z.; Jin, T.; Huang, L.; Feng, L.*
Retrosynthetic construction of protein-mimetic high-coordination ion pockets for rare-earth separation
Submitted, **2026**

14. Cho, E.;† **Lee, J.**;† Lee, S.; Yoon, J. H.; Jang, S.; Kim, S.; Choe, W.*
Design and Synthesis of Merged Net Porphyrinic Metal-Organic Frameworks
Manuscript in preparation, **2026**

15. Ju, E.;† **Lee, J.**;† Nam, J.; Park, Y. S.;* Choe, W.*
Vacancy Control of Metal-Organic Frameworks
Manuscript in preparation, **2026**

Awards & Honors

06/2026	Best Poster Award, Inorganic Chemistry Division, Korean Chemical Society
11/2023	Best Poster Award, Korean Crystallographic Association
05/2023	UBSI Basic Science Researcher Incubating Fellowship Program, UNIST

06/2022 Best Poster Award, Inorganic Chemistry Division, Korean Chemical Society

11/2019 Best Poster Award, Korean Crystallographic Association

Patents

1. "Nonplanarity porphyrinic metal-organic frameworks and method for preparing same", Korea Patent Registration Number 10-2263645 (06/04/2021)
2. "Design and synthesis of porphyrinic metal-organic frameworks with pyridyl ligand", Korea Patent Registration Number 10-2171468 (10/23/2020)
3. "Porphyrinic metal-organic frameworks having multiple porphyrin ligands and its preparing method", Korea Patent Registration Number 10-2166662 (10/12/2020)
4. "Nanographene-Incorporated Metal-Organic Frameworks", Korea Patent Application Number (10-2024-0118025 (08/30/2024)

Conference Presentations

- **2026 KCS Inorganic Chemistry Division Summer Symposium**, Functional Confined Space in Metal-Organic Frameworks: Toward Molecular Growth and Selective Capture, Republic of Korea (Jun. 2026, Poster)
- **MOF 2026 10th International Conference**, Rational Synthetic Approaches of Porphyrin-based Metal–Organic Frameworks for Advanced Functionality, USA (May. 2026, Poster)
- **Gordon Research Conference (GRC, 2025)**, Pore Space Engineering of Metal-Organic Frameworks for CO_2 Capture, USA (Aug. 2025, Poster)
- **133rd General Meeting of the Korean Chemical Society**, Rational Synthetic Strategy for Porphyrinic and Related Metal-Organic Frameworks Targeting Advanced Functionality, Republic of Korea (Apr. 2024, Oral)
- **2023 Korean Crystallographic Association**, Unveiling Porphyrin-based Merged-Net Metal-Organic Frameworks and Systematic Pore Engineering, Republic of Korea (Nov. 2023, Poster)
- **Gordon Research Conference (GRC, 2023)**, Unveiling Porphyrin-based Merged-Net Metal-Organic Frameworks and Systematic Pore Engineering, USA (Aug. 2023, Poster)
- **2023 KCS Inorganic Chemistry Division Summer Symposium**, Engineering Energy Transfer Pathways in Nanographene-based Metal-Organic Frameworks by Inhibiting Charge Transfer, Republic of Korea (Jun. 2023, Poster)
- **2022 KCS Inorganic Chemistry Division Summer Symposium**, Nanozyme based on Porphyrinic Metal-Organic Framework for Electrocatalytic CO_2 Reduction, Republic of Korea (Jun. 2022, Poster)
- **129th General Meeting of the Korean Chemical Society**, Nanographene-based Metal-Organic Frameworks, Republic of Korea (Apr. 2022, Poster)
- **AsCA2019, 16th Conference of the Asia Crystallographic Association**, Non-Planarity in 2D Porphyrinic Metal-Organic Framework, Singapore (Dec. 2019, Oral)
- **2019 Korean Crystallographic Association**, Non-planarity in 2D Porphyrinic Metal-Organic Framework, Republic of Korea (Nov. 2019, Poster)
- **5th Japan-Korea Joint Symposium on Hydrogen in Materials**, Non-Planarity in 2D Porphyrinic Metal-Organic Framework, Japan (Nov. 2019, Poster)
- **124th General Meeting of the Korean Chemistry Society**, Non-Planarity in 2D Porphyrinic Metal-Organic Framework, Republic of Korea (Oct. 2019, Poster)

- **2018 KCS Inorganic Chemistry Division Summer Symposium**, Metal-organic framework based on polybenzene, Republic of Korea (Jun. 2018, Poster)
- **MOF 2018 6th International Conference on Metal-Organic Frameworks & Open Framework Compounds**, Non-Planarity in 2D Porphyrinic Metal-Organic Framework, New Zealand (Dec. 2018, Poster)
- **2017 KCS Inorganic Chemistry Division Summer Symposium**, Curved Surface in Porphyrinic Metal-Organic Frameworks, Republic of Korea (Jul. 2017, Poster)
- **2016 KCS Inorganic Chemistry Division Summer Symposium**, Design and Synthesis of Mesoporous Porphyrinic Metal-Organic Frameworks Towards Biomimetic Catalysis, Republic of Korea (Jul. 2016, Poster)
- **2015 Korean Chemical Society Yeongnam Regional Division Symposium**, Design and Synthesis of Mesoporous Porphyrinic Metal-Organic Frameworks Towards Biomimetic Catalysis, Republic of Korea (Aug. 2015, Poster)

Professional Activities

Research Internship	**U-SURF (UNIST-Summer Undergraduate Research Fellowship)** (Prof. Wonyoung Choe, Jul. 2014)
Volunteer Activities	**Preparatory Committee of Laboratory Movement** New laboratory design and laboratory movement preparation committee activities (Dec. 2015 – Jan. 2016)

References

Prof. Wonyoung Choe (Ph.D. supervisor)

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